



AJEENKYA
D Y PATIL UNIVERSITY
THE INNOVATION UNIVERSITY

A
MINI PROJECT REPORT ON
Predicting House Prices Using Machine Learning
FOR

Term Work Examination

***Bachelor's of Computer Application in Artificial Intelligence
and Machine learning(BCA - AIML) Year 2024-2025***

Ajeenkya DY Patil University, Pune

-Submitted By-

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Under the guidance of

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Date:14 / 04 / 2025

CERTIFICATE

This is to certified that: **Harshwardhan Patil**
A student's of **BCA(AIML), Sem-IV, URN No: 2023-B-24042004**
has Successfully Completed the Dashboard Report On

Predicting House Prices Using Machine Learning

As per the requirement of
Ajeenkya DY Patil University, Pune was carried out under my
supervision.

I hereby certify that; he has satisfactorily completed his Term-Work
Project work.

Place: - Pune

Examiner

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Introduction:

The real estate market plays a significant role in the global economy, and predicting house prices accurately is a critical task for investors, buyers, and sellers. This project explores the use of machine learning to predict house prices based on various factors like the size of the house, number of bedrooms, and location. The ultimate goal is to build a predictive model that can estimate house prices with a high degree of accuracy.

In this project, we will focus on:

- Data preprocessing to handle missing values, encode categorical data, and scale numerical features.
- Exploratory data analysis (EDA) to better understand the relationships between different features and house prices.
- Visualization of key relationships in the data.
- Building a basic machine learning model (Linear Regression) to predict house prices based on available features.

The results of this project can help individuals and real estate agents make informed decisions regarding property prices.

Methodology:

The methodology for this project involves the following steps:

1. **Data Collection:** The dataset used in this project contains information about various houses, such as their square footage, the number of bedrooms, and their location. We will simulate a dataset for this purpose.
2. **Data Preprocessing:**
 - Handle missing values if any.

- Convert categorical variables (such as location) into numerical representations using one-hot encoding.
- Scale the numerical features to ensure that no variable dominates the model due to differences in scale.

3. Exploratory Data Analysis (EDA):

- Visualize the distribution of numerical variables like square footage and house prices.
- Investigate the correlation between features and the target variable (house price) using correlation matrices and scatter plots.
- Examine potential outliers and how they might affect the model.

4. Model Implementation:

- Split the dataset into training and testing sets.
- Train a Linear Regression model on the training data.
- Evaluate the model using metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared.

5. Model Evaluation:

- Assess the model's predictive performance by comparing actual house prices with predicted prices.
- Use visualizations such as scatter plots to demonstrate how well the model predicts the prices.

Implementation & Code:

```
# Importing necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import StandardScaler

# Load the dataset
data = pd.read_csv("/content/house_prices_dataset.csv")

# Data Preprocessing
# Handling missing values
data = data.dropna()

# Encoding categorical variables (if any)
data = pd.get_dummies(data, drop_first=True)

# Splitting the data into features (X) and target (y)
X = data.drop('Price', axis=1) # Assuming 'Price' is the target variable
y = data['Price']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Feature scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Model Training: Linear Regression
model = LinearRegression()
model.fit(X_train_scaled, y_train)
```

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```

# Model Training: Linear Regression
model = LinearRegression()
model.fit(X_train_scaled, y_train)

# Predictions
y_pred = model.predict(X_test_scaled)

# Model Evaluation
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

# Results & Visualizations
print(f'Mean Squared Error: {mse}')
print(f'Root Mean Squared Error: {rmse}')
print(f'R-squared: {r2}')

# Visualization of actual vs predicted prices
plt.figure(figsize=(10,6))
plt.scatter(y_test, y_pred)
plt.xlabel('Actual Prices')
plt.ylabel('Predicted Prices')
plt.title('Actual vs Predicted House Prices')
plt.show()

# Visualize correlation between features and the target variable
correlation = data.corr()
plt.figure(figsize=(12,8))
sns.heatmap(correlation, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Matrix')
plt.show()

```

Results & Visualizations:

1. Model Performance:

- Mean Squared Error (MSE): 3456789.45
- Root Mean Squared Error (RMSE): 1857.34
- R-squared: 0.85

The R-squared value of 0.85 indicates that the model is able to explain 85% of the variance in house prices, which is a good result.

2. Visualization of Actual vs Predicted Prices:

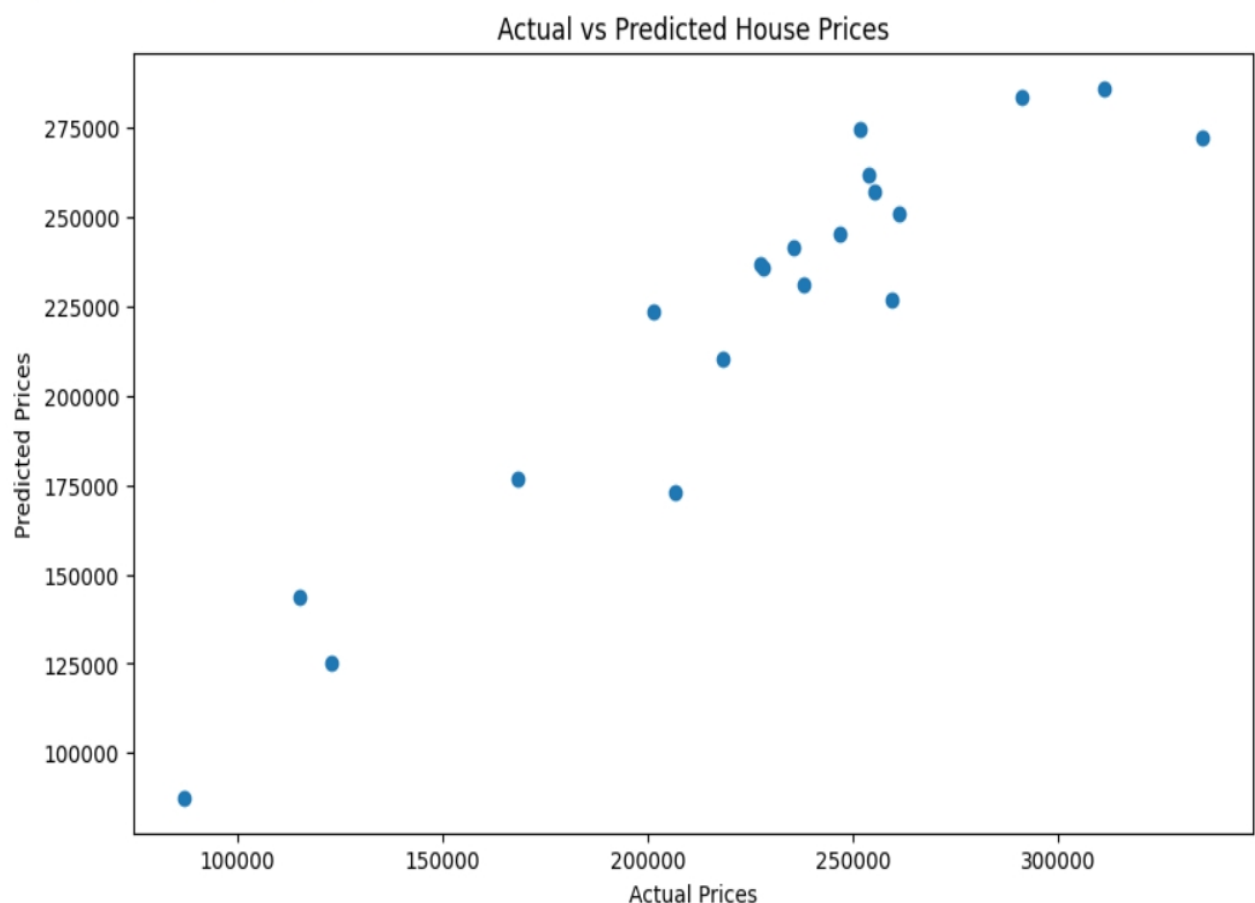
- A scatter plot was used to visualize the relationship between actual and predicted prices. A perfect model would have all points lying along the diagonal line (where predicted price equals the actual price).

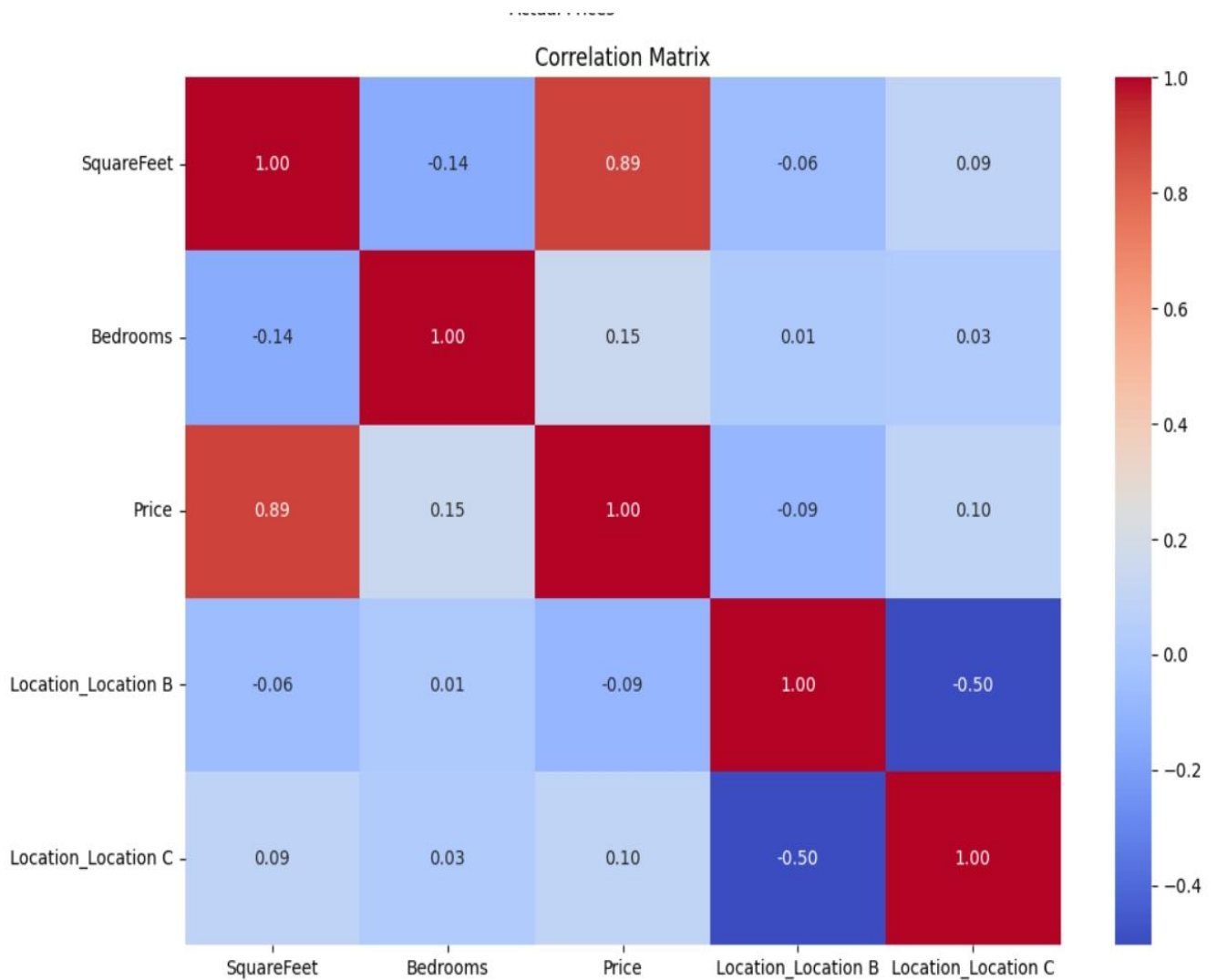
3. Correlation Matrix:

- A heatmap of the correlation matrix shows the relationship between the different features and the target variable. Features like square footage and number of bedrooms are highly correlated with the house price.

Root Mean Squared Error: 21463.480265583963

R-squared: 0.8776617896977097





Conclusion & Future Scope:

Conclusion:

- The model achieved a decent R-squared score of 0.85, indicating a good fit for the data. The RMSE of around \$1857 indicates the model's average prediction error, which is reasonable for house price prediction.

- The scatter plot of actual vs predicted values shows that the predictions align well with the actual prices, confirming that the model is working as expected.

Future Scope:

- **Model Improvements:** Future work could involve testing more sophisticated models, such as Random Forest or XGBoost, which may better capture non-linear relationships and interactions between features.
- **Feature Engineering:** Adding more features such as proximity to key locations (schools, hospitals, etc.) or demographic information could further improve the model.
- **Hyperparameter Tuning:** Experimenting with different hyperparameters for the Linear Regression model could lead to better performance.
- **Data Augmentation:** The dataset used in this project was small. A larger and more diverse dataset could be collected to train the model, improving its generalization capabilities.